

Jesse Anderson

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Summary

Data Systems and Automation Engineer with Master's in CS and Bachelor's in Chemical Engineering specializing in ETL pipelines, embedded IoT, and full stack development. Published researcher with 6 peer reviewed coauthored papers in computational biology and image analysis.

Education

University of Illinois at Urbana-Champaign | Master of Computer Science

GPA: 4.00 | Courses: Cloud, ML, Distributed Systems, Scientific Visualization, Stats, DB, Data Cleaning

University of Illinois at Chicago | BS Chemical Engineering

GPA: 3.77 | Concentrations: BioChemE & Process Automation | Minor: Math & Computer Science

Technical Skills

Engineering & Simulation: Process Automation, Aspen Plus, SuperPro, Blueprint Reading, Circuit Design, IoT.

Languages & Compute: Python, Rust, C++, MATLAB, SQL (PostgreSQL/DuckDB), CUDA, VBA, R.

Data & Embedded: PyTorch, OpenCV, ETL Pipelines, Raspberry Pi, ESP32, Docker, Scikit-learn, Pandas.

Web & Deployment: WebAssembly (WASM), Flask, JS, React, GitHub Actions, AWS/GCP, Linux/Bash.

AI & LLM Engineering: Agentic orchestration, structured prompting, tool/function calling, eval/regression harnesses

Experience

Engineer | G5 Environmental | Chicago, IL

September 2024-Present

- Architect and deploy embedded sensor modules (ESP32, pressure, RPM, GPS) to optimize fleet reliability.
- Develop automated Python testing workflows (pytest) and HIL validation scripts to verify system behavior.
- Diagnose hydraulic leaks and electrical faults by analyzing sensor data logs, reducing diagnostic time.

Engineer | UL Solutions | Northbrook, IL

June 2023-April 2024

- Design automation software in Python and VBA to eliminate 500+ hours of tedious workflows annually.
- Develop Python software with OpenCV/PyMuPDF to automatically detect changes in CAD files.
- Establish ETL pipeline to reduce redundant analytical tasks by 400 hours per year.
- Build C#/NET/VBA tools to modernize legacy workflows, codifying SME knowledge into repeatable processes.

Research Assistant | University of Illinois at Chicago | Chicago, IL

December 2020-June 2023

- Co-authored 6 peer-reviewed papers (e.g., Scientific Reports, Biophysical Journal) [[Publications](#)].

Engineering Intern | United Conveyor Corporation | Waukegan, IL

May 2022-August 2022

Projects

Engineering Tools Suite | JavaScript, WASM, DBSCAN | tools.jesse-anderson.net

- Developed a suite of web-based engineering utilities, leveraging WebAssembly for client-side calculation.

linreg-core Rust Crate | Rust, WebAssembly, Python, VBA | crates.io/crates/linreg-core | [Writeup](#)

- Linear regression library with OLS, WLS, Ridge, Lasso, and comprehensive diagnostic testing.
- Bindings for WASM(wasm-bindgen), Python(PyO3), and VBA(FFI dll); validated against R / Python libraries.

Oak Park Crime Tracker | Python, DuckDB (WASM), Raspberry Pi | opcrime.jesse-anderson.net | [Writeup](#)

- Built static web app with daily ETL pipeline on RPi; uses client-side DuckDB for serverless SQL queries.

Oak Park Data Science Dashboard | Python, UMAP, ML, NLP | <https://opcrimedts.jesse-anderson.net/> | [Writeup](#)

- Daily statistical analysis of crimes reported by Oak Park PD; spatiotemporal, clustered, and per zone analysis.

Industrial IoT Sensor Network | ESP32, Python, PostgreSQL, MQTT | [Writeup](#)

- Engineered fault-tolerant monitoring systems (SCD41/PMS7003) pushing real time data to redundant SQL DBs.

N-Back Cognitive Training | Rust, WebAssembly, JavaScript | nback.jesse-anderson.net

- Developed n-back memory training game using Rust compiled to WASM for high-performance browser execution.

Senior Design: Bio-Process Simulation | Aspen Plus, SuperPro, Process Design | [Writeup](#)